18/3/26

Q1] Evaluate $\int_0^\infty \frac{x^3}{3^x}\,dx$

$\rightarrow$ $I = \int_0^\infty \frac{x^3}{3^x}\,dx = \int_0^\infty x^3 e^{-x\ln 3}\,dx$

use gamma integral

$\int_0^\infty x^n e^{-ax}\,dx = \frac{n!}{a^{n+1}}$

$n = 3$; $a = \ln 3$

$I = \frac{3!}{(\ln 3)^4} = \frac{6}{(\ln 3)^4}$

Q(2) Evaluate $\int_0^\infty x^n e^{-x^m}\,dx$

Let, $t = x^m \Rightarrow x = t^{1/m}$

$dx = \frac{1}{m} t^{1/m - 1}\,dt$

$I = \int_0^\infty t^{n/m} e^{-t} \frac{1}{m} t^{\frac{1}{m}-1}\,dt$

$= \frac{1}{m}\int_0^\infty t^{\frac{n+1}{m}-1} e^{-t}\,dt$

using gamma function,

$\Gamma k = \int_0^\infty t^{k-1} e^{-t}\,dt$

$I = \frac{1}{m}\Gamma\frac{n+1}{m}$

Q3 Express $\int_{-1}^{1} (1+x)^m (1-x)^n\,dx$ in terms of gamma function

$I = \int_{-1}^{1} (1+x)^m (1-x)^n\,dx$

put $x = 1-2t$; $dx = -2dt$

$x=-1 \Rightarrow t=1$, , $x=1 \rightarrow t=0$

$$I = 2^{m+n+1} \int_0^1 t^n (1-t)^m dt$$

using beta function,

$$B(p,q) = \int_0^1 t^{p-1}(1-t)^{q-1} dt = \frac{\Gamma p \, \Gamma q}{\Gamma(p+q)}$$

$$I = 2^{m+n+1} \frac{\Gamma m+1 \, \Gamma n+1}{\Gamma m+n+2}$$

Q4] show that $\int_0^\infty \frac{x^{n-1}}{(a+bx)^{m+n}} dx = \frac{1}{a^m b^n}$

$$I = \int_0^\infty \frac{x^{m-1}}{(a+bx)^{m+n}} dx$$

putting $x = \frac{a}{b} t$, $dx = \frac{a}{b} dt$

$$I = \frac{a^{-n}}{b^m} \int_0^\infty \frac{t^{m-1}}{(1+t)^{m+n}} dt$$

using beta function identity

$$\int_0^\infty \frac{t^{m-1}}{(1+t)^{m+n}} dt = B(m,n)$$

$$I = \frac{1}{a^n b^m} B(m,n)$$

hence prooved

Q5) prove that $\int_0^\infty e^{-\infty} \frac{\sin x}{x} dx = \cot^{-1} a$

$I(a) = \int_0^\infty e^{-ax} \frac{\sin x}{x} dx \; (I)$

$I'(a) = \frac{d}{da} \int_0^\infty e^{-ax} \frac{\sin x}{x} dx$

$= \int_0^\infty \frac{\partial}{\partial a} e^{-ax} \frac{\sin x}{x} dx$

$= \int_0^\infty -x\, e^{-ax} \frac{\sin x}{x} dx$

standard result

$= -\int_0^\infty e^{-ax} \sin x \, dx$

$= \frac{-e^{-ax}}{(-a)^2 + b^2} [a \sin bx - b \cos bx]$

$= -\left[\frac{e^{-ax}}{a^2+1}\right] (-a \sin x - \cos x) \Big]_0^\infty$

$= -\left[\frac{1}{a^2+1} (-1)\right]$

$I'(a) = \frac{-1}{a^2+1}$

$\frac{dI}{da} = \frac{-1}{a^2+1}$

$$dI = \int \frac{-1}{a^2+1}\, da$$

$$= -\tan^{-1}(a) + c \quad - (\mathrm{II})$$

Put (∞) at the place of 'a' the equation 10 it will become

$$\frac{e^{-\infty x} \sin x}{x}\, dx = 0$$

put The value in (II)

$$= -\tan^{-1}(\infty) + c$$

$$= -\frac{\pi}{2} + c$$

$$\left[c = \frac{\pi}{2}\right]$$ so equation 2 becomes

$$I(a) = \frac{\pi}{2} - \tan^{-1}(a)$$

$$I(a) = \cot^{-1}(a)$$

hence proved

$$\int_0^\infty \frac{\sin x}{x}\, dx = \frac{\pi}{2}$$ this can posible if $a = 0$

$$\int_0^\infty e^{-ax} \frac{\sin x}{x}\, dx$$

$\therefore$ $a = 0$

so, $\int e^{-ax} \frac{\sin x}{x}\, dx = \cot^{-1}(0) = \frac{\pi}{2}$